Title: FLIGHT MANAGEMENT SYSTEM USING C++

1. Problem Statement

To develop a Flight Management System using C++ to manage flight details and passenger bookings. The system stores information such as Flight Number, Source, Destination, Passenger Name, and Ticket Number. It allows users to book tickets, display flight details, and manage passenger information using suitable data structures.

2. Objectives

- To understand the concept of Singly Linked List.
- To implement dynamic memory allocation in C++.
- To store and manage flight details efficiently..
- To implement passenger ticket booking functionality.
- To display flight and passenger information.
- To improve programming and problem-solving skills.

3. Algorithm

Step 1: Start the program.

Step 2: Define a structure `Flight` containing:

- Flight Number
- Source
- Destination
- Passenger Name
- Ticket Number
- Pointer `next` (if using a linked list)

Step 3: Initialize `head = NULL` if a linked list is used.

Step 4: Display the menu with the following options:

1. Add Flight Details
2. Book Flight Ticket
3. Display Flight Details
4. Display Passenger Details
5. Exit

Step 5: Accept the user's choice.

Step 6: If the user selects Add Flight Details, accept the flight information and store it.

Step 7: If the user selects Book Flight Ticket, accept the passenger details and assign a ticket number.

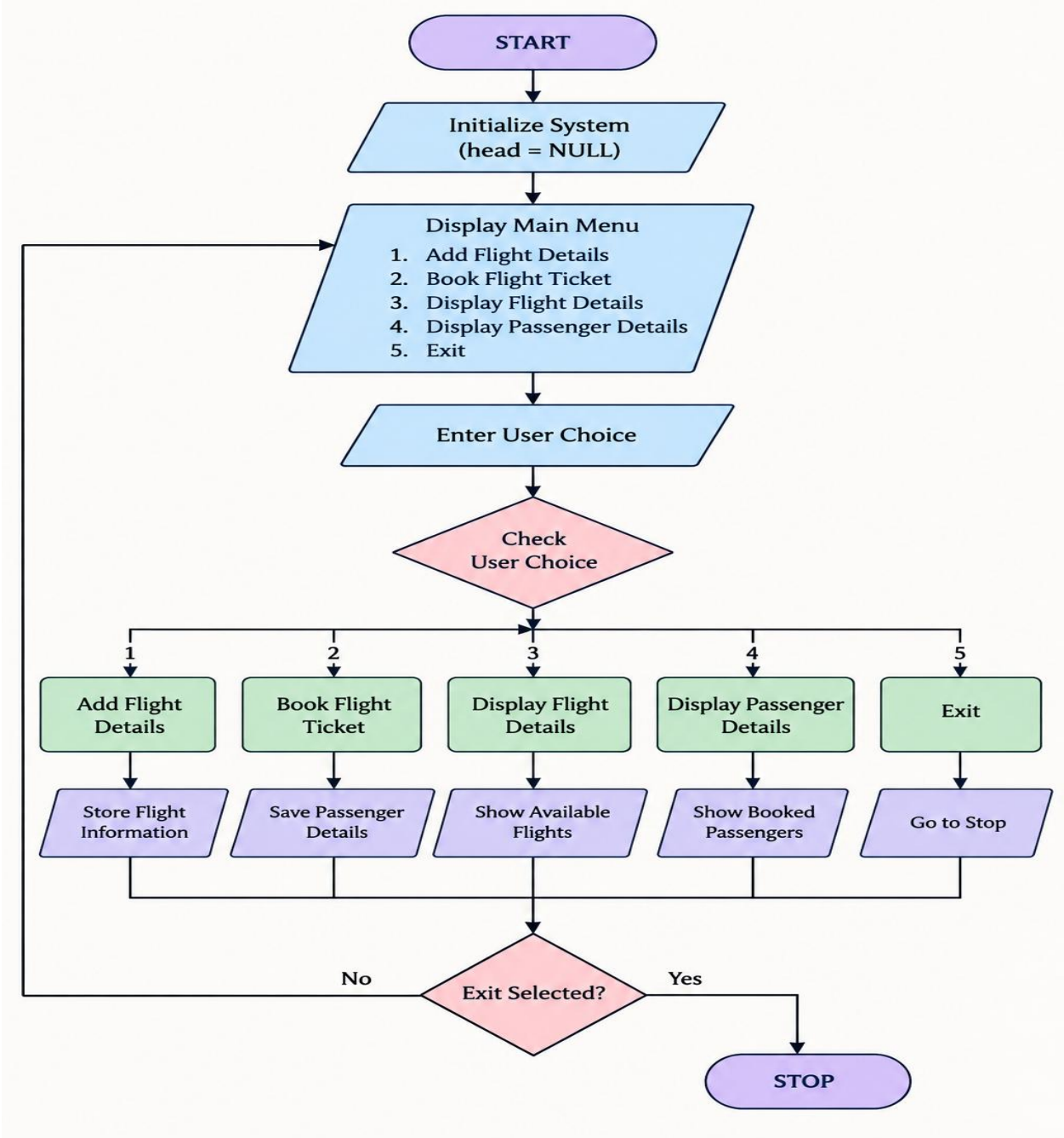
Step 8: If the user selects Display Flight Details, display the available flight information.

Step 9: If the user selects Display Passenger Details, display the booked passenger information.

Step 10: Repeat the menu until the user selects Exit.

Step 11: Stop the program.

4. Flowchart



5. Conclusion

The Flight Management System using C++ demonstrates the practical application of data structures in managing flight and passenger information. It provides basic functionalities for adding flight details, booking tickets, and displaying records. This project helps improve understanding of C++ programming, data management, and problem-solving skills.